Separation of arguments: consequences for dependents of nouns

Veronika Gvozdovaitė veronika.g@nyu.edu **New York University**

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Takeaways

- An argument for separation from Schein (2017) also applies (i) to relational nouns and (ii) to deverbal nominalizations
- Relational nouns then cannot denote two-place predicates: their apparent dependent is separated
- Complex event nominals (CENs) cannot have a Theme argument introduced in the nominal layer

0. Schein's (2017) separation argument: modal adverbs

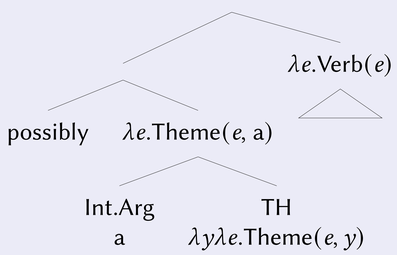
- If an event predicate and a thematic role predicate are introduced in distinct parts of a derivation, then they can interact separately with operators: a *semantic wedge* between the two predicates Williams 2015
- There are such cases where an operator has in its scope only the thematic argument, and not the event predicate. As such, the argument is separated from the event predicate, structurally Schein 2017, see also Williams 2015
- The meaning of (1) is like (2): the modal Adv *possibly* has in its scope only the thematic role predicate (Theme), but not the event predicate (Gather) Schein 2017

(1) The cops gathered the students and possibly the professors.

(2) 'There is an event e that is a gathering; the cops are agents in e; the students are participants in e; and possibly the professors are participants in e as well.' adapted from Williams 2015:192-193

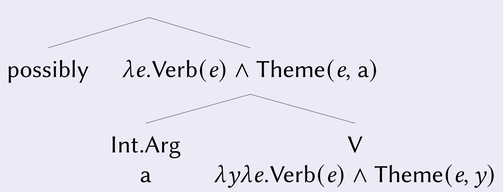
- *possibly* only has $\lambda e.\text{Theme}(e, \text{the-profs})$ in its scope
- If a separate node introduces the thematic argument, *possibly* can combine with $\lambda e.\text{Theme}(e, \text{the-profs})$ to the exclusion of the event predicate

(3)



- If the V node introduces both the event predicate and a Theme argument, then *possibly* will combine with the entire lexical meaning of V and cannot penetrate it to target only the Theme predicate, making (2) impossible

(4)



The Theme predicate is introduced in the derivation separately from the event predicate.

1. Projectionist nouns #1: Relational nouns as optionally transitive predicates

- Some nouns denote relations over pairs of entities, rather than just a set of entities: *relational nouns* Barker (1991:68-81)
- *Edge* entails the existence of an entity that has an edge, *brother* the existence of a sibling, *king* – the existence of some organization, *picture* – something that is represented in the picture etc.

(5) $[_{DP}$ The king of Denmark] is tall.
$[\![\text{king}]\!] = \lambda x\lambda y.\text{King}(x, y)$
$[\![\text{the king of Denmark}]\!] = \iota x.\text{King}(x, \text{denmark})$

- The *of*-PP is optional, but the inference survives with a bare noun, in some proposals encoded via existential binding of the argument variable Barker (2011)

(6) $[_{DP}$ The king] is tall.
$\rightsquigarrow$ the king of some place
$[\![\text{the king}]\!] = \iota x\exists y.\text{King}(x, y)$

- Cf. optionally transitive verbs like *eat*, which have an implied argument in their intransitive use Dowty 1981 *apud* Gillon 2012

(7) Amy ate breakfast.
$[\![\text{Amy eat breakfast}]\!] = \text{Eat}(\text{amy}, \text{breakfast})$

(8) Amy ate.
$\rightsquigarrow$ Amy ate some thing or other
$[\![\text{Amy eat}]\!] = \exists y.\text{Eat}(\text{amy}, y)$

- In both cases, the predicate and the argument(s) are introduced together, by one lexical node (head noun *king* / head verb *eat*) in the derivation
  - (a projectionist denotation for relational nouns)
- The argument carries over to conjoint representations too, if the meaning parts are introduced at a single point in the derivation:

(9) $[\![\text{eat}]\!] = \lambda x\lambda y\lambda e.\text{Eat}(e) \wedge \text{Theme}(e, y)$

(10) $[\![\text{king}]\!] = \lambda y\lambda x.\text{King}(x) \wedge \text{Rel}(x, y)$

2. Separating the argument from the relational noun

- Here's Schein's *possibly* argument applied to relational nouns, *mutatis mutandis*

(11) the picture of the students and possibly (of) the professors

(12) the king of England and possibly the Chagos Archipelago

- (11) can express a thought like (13)/(14)

(13) 'Either the picture includes the students, or the picture includes the students and the professors'

(14) 'There is a picture x, and the students are representations in x, and possibly the professors are also representations in x'

- *possibly* does not have $\lambda x.\text{picture}(x)$ in its scope, only the representation argument
- Therefore, $\lambda x.\text{picture}(x)$ and its apparent dependent are introduced into the derivation separately

The two entities of relational nouns have to be introduced separately in the derivation, not together *pace* Barker 1991 et seq. but with Adger 2013

3. Projectionist nouns #2: deverbal nominalizations

- Similarly for *possibly* with complex event nominalizations (CENs)

(15) the (ongoing) surrounding of the science building and possibly (of) the library

(16) the cops' gathering of the students and possibly the professors in 10 minutes

(17) the transmission of the latest news and possibly a secret message

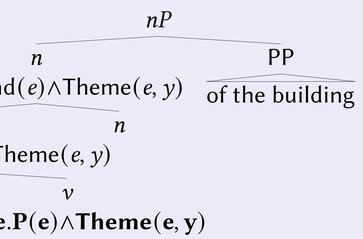
- Assuming (15) can express a thought like (18)/(19), the argument for separation goes through for CENs as well

(18) Either the science building is surrounded, or the science building and the library are surrounded

(19) There is an e that is a surrounding; the science building is a participant/affectee in e, and possibly the library is a participant/affectee in e as well

- This presents a problem for the allosemy approach to CENs (Wood 2023)
- In this approach, heads combine before arguments: the phrasal dependent (*science building*) is syntactically introduced only in the nominal layer
- Yet, its thematic role (Theme) is introduced by v inside the complex head, together with the event predicate

(20)



- The allosemic account depends on this denotation of v: it is what makes the structure in (20) an event nominalization (CEN), not a result/referential one
- Yet this projectionist denotation of v is incompatible with (15)-(17), which require separation

The thematic relation has to be introduced together with the nominal dependent in the syntax. If the dependent NP is in the nominal layer, then its thematic relation also has to be nominal *pace* Wood (2023)

Conclusions

- Schein's argument for separation using *possibly* also applies to:

(i) relational nouns and their dependents (*king of Denmark*), suggesting that their argument is introduced by a separate thematic piece

(21) $\lambda x.\text{King}(x) \wedge \text{Rel}(x, \text{denmark})$

(ii) complex event nominals, which weakens the complex-head story for nominalization ambiguity

- In CENs, either the nominal argument bears a Theme relation and is introduced in the verbal layers of structure, or it bears a nominal thematic role (like e.g. the dependent of a relational noun) and is introduced in the nominal layer of structure

Selected references

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